

DONGHYUN SOHN

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EDUCATION

Northwestern University

Evanston, IL, United States

Ph.D. in Computer Science

Sep. 2022 - Present

- Advisor: Jennie Rogers

Sogang University

Seoul, Republic of Korea

B.S. in Computer Science and Engineering

Mar. 2016 - Aug. 2021

- Summa Cum Laude (2 years of military service included)

EXPERIENCE

Graduate Research Assistant, Northwestern University

Evanston, IL, United States

Sep. 2022 - Present

- Introduced Alchemy, a protocol-agnostic optimization framework for privacy-preserving query processing, combining traditional optimization techniques with circuit-aware cost modeling. Achieved up to $100\times$ speedups in secure query execution across multiple cryptographic protocols.
- Built HAMMER, a hybrid FHE+MPC columnar query engine partitioning SQL operators across plaintext, BFV, and MPC domains. Designed a novel FHE-to-MPC conversion protocol and parallelized the pipeline across CPU threads and GPU, running TPC-H queries at SF=1 in under a minute. **SIGMOD 2027 (under review)**
- Built a Differential Privacy-compliant data synthesis framework for OLAP performance regression, ensuring high fidelity simulation of statistical distributions and physical storage layouts in DuckDB/Parquet.
- Investigating LLM-based NL2SQL translation with a focus on schema incompleteness as a structural bottleneck; evaluating retrieval-augmented approaches on the BIRD benchmark.
- Designed a cost-aware optimizer for secure SQL workloads with ML-based runtime prediction, incorporating AWS hardware characteristics for performance- and cost-efficient instance selection.

Undergraduate Research Intern, Sogang University

Seoul, Republic of Korea

Aug. 2020 - Dec. 2021

- Optimized the BIRCH clustering algorithm by designing a disk-based Clustering Feature tree with selective buffering, reducing disk I/O and improving clustering quality by 7% for large-scale datasets.

SELECTED PUBLICATIONS

1. **Donghyun Sohn**, Jennie Rogers, “ScanTwin: Simulating Performance Regressions Without Access to Tenant Data,” *SeQureDB Workshop at SIGMOD*, 2026. [Paper]
2. **Donghyun Sohn**, Kelly Jiang, Nicolas Hammer, Jennie Rogers, “Alchemy: A Query Optimization Framework for Oblivious SQL,” *Proceedings of the VLDB Endowment*, 2025. [Paper] [Code]
3. **Donghyun Sohn**, Xiling Li, Jennie Rogers, “Everything You Always Wanted to Know About Secure and Private Database Systems (but were Afraid to Ask),” *Data Engineering Bulletin*, 2023. [Paper]
4. **Donghyun Sohn**, Sungwon Jung, “Disk-based BACF Tree for Clustering Massive Datasets,” *Korea Software Congress*, 2021. [Paper] [Code]

SERVICE

Program Committee Member

SIGMOD Availability & Reproducibility Initiative (ARI)

2024, 2025

SKILLS

Programming: C, C++, Java, Python, SQL

Systems/Tools: AWS, Docker, Ubuntu, EMP-Toolkit, OpenFHE